

QuantumShield CyberDefense Technical Audit Report

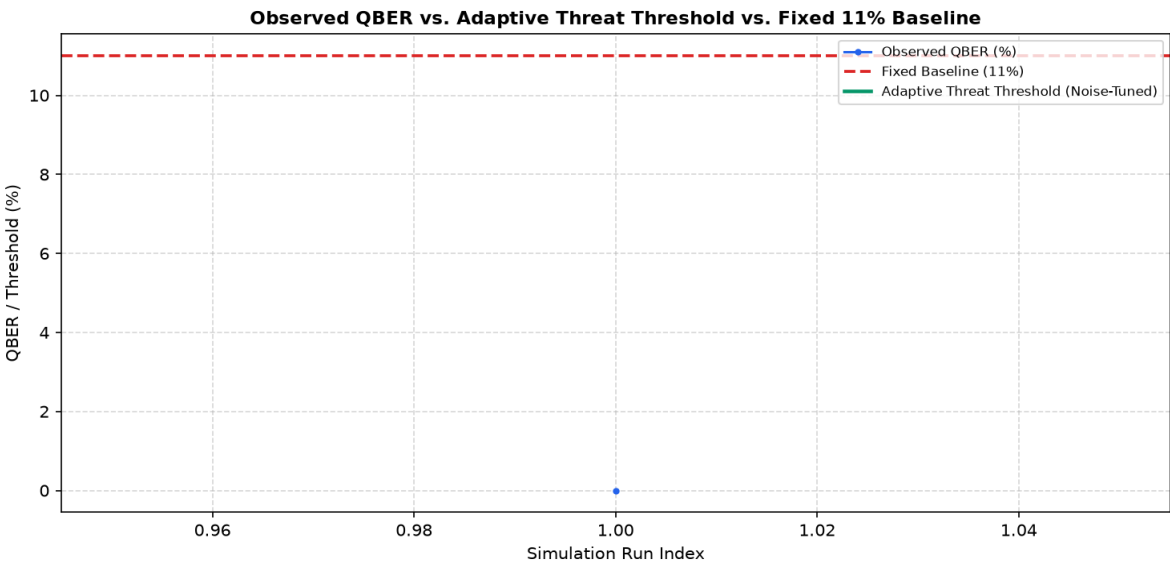
BB84 Quantum Key Distribution & RSA-2048 Digital Signature Security

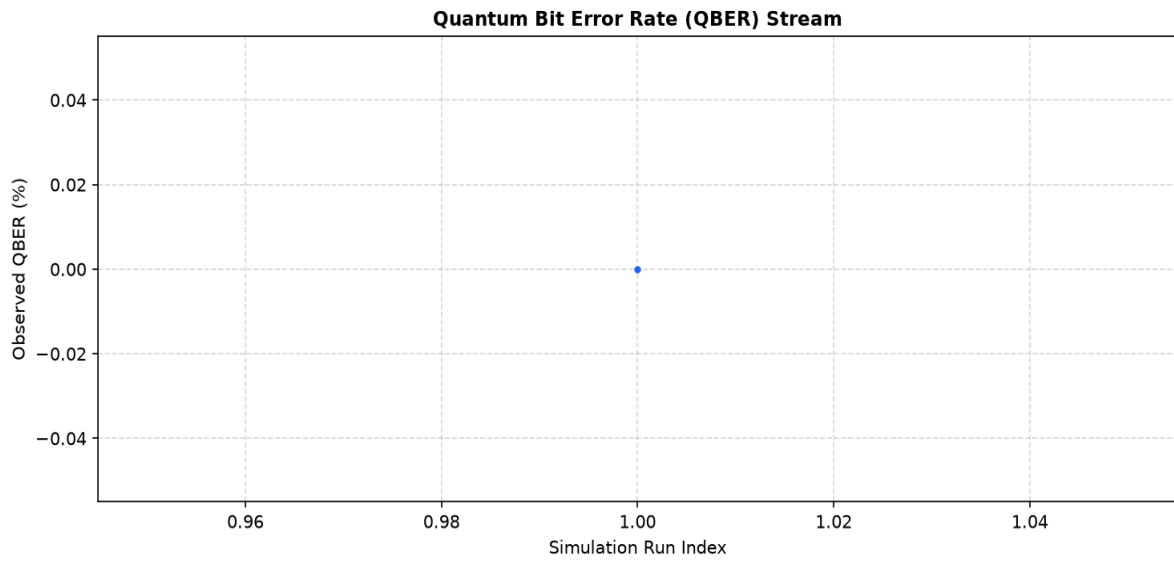
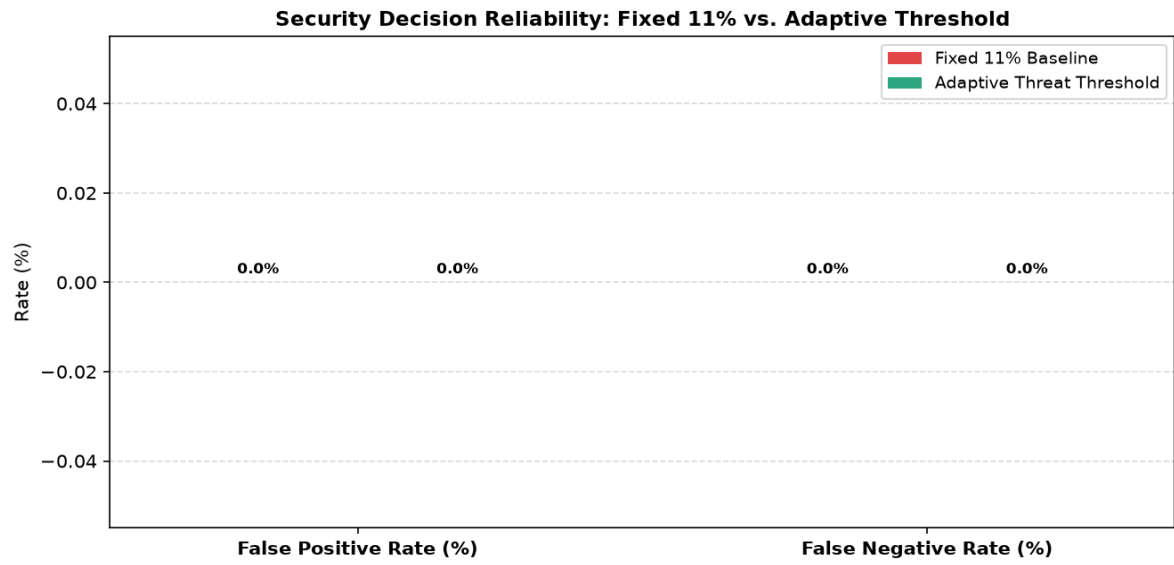
Scenario Context: Case 1: Secure Communication | Generated on: 2026-09-15 22:12:05

1. Executive Performance Summary

Metric Parameter	Value	Notes / Evaluation
Total Simulation Runs	1	Monte Carlo BB84 iterations
Average Observed QBER	0.00%	Min: 0.0%, Max: 0.0%
Traditional Fixed Threshold	11.00%	Static BB84 theoretical limit
Adaptive Threat Threshold (Avg)	11.00%	Dynamically adjusted to channel noise
Fixed Threshold False Positive Rate	0.0%	False attack alerts under high noise
Adaptive Threat False Positive Rate	0.0%	Eliminated false alarms
Avg Key Generation Time	48.02 ms	Sifted key agreement latency
Avg Sifted Key Length	17.0 bits	Secret key output size

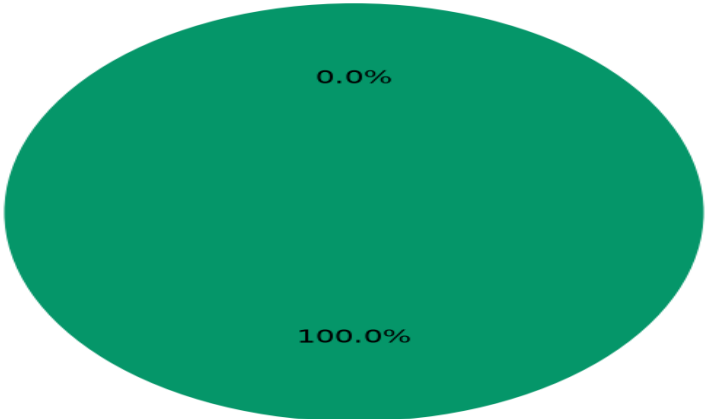
2. Graphical Telemetry Analysis





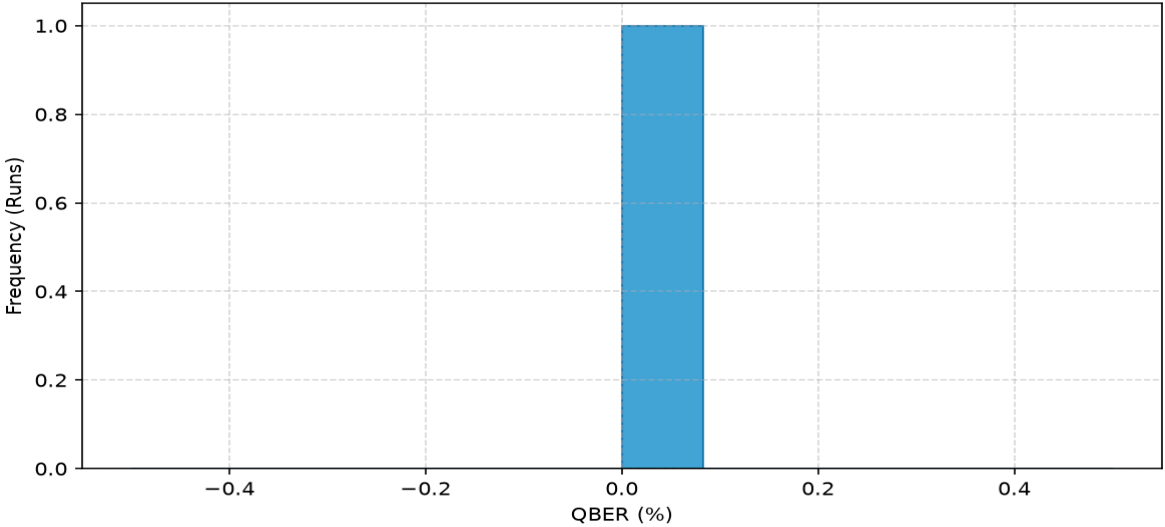
Key Security Decision Ratio

Rejected / Eve (0)



Accepted (1)

QBER Distribution Density



3. Security Analysis & Adaptive Threat Observation

SECURE CHANNEL CONFIRMED: The communication channel is secure. The observed QBER is well within the Adaptive Threat Threshold. Generated keys are approved for AES-256 confidential enterprise payload encryption.